

**PAPER\_1617 — Dark Energy Density  $\Omega_\Lambda = \Omega_\Lambda = SSQ + F \cdot SSQ + F^2 \cdot D\_BSFG - F^2 \cdot SSQ^2 \approx 0.6838$**

**Author:** Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Cosmology **Date:** June 18, 2026 **Status:** CLOSED — 0.182% closure

**Observation**

PAPER\_1209GG\_S644 (Cosmology Unified Proof Set) gives the formula:

$\Omega_\Lambda = SSQ + F \cdot SSQ + F^2 \cdot D\_BSFG - F^2 \cdot SSQ^2 \approx 0.6838$

Residual: **0.182%**.

**UQFF Closed Identity**

$\Omega_\Lambda = SSQ + F \cdot SSQ + F^2 \cdot D\_BSFG - F^2 \cdot SSQ^2 \approx 0.6838 \quad 0.182\%$

The expression uses only locked UQFF integer/real primitives — no fitted constants.

**$\Lambda$ CDM Standard Cosmology Coverage**

With this catch-up, **all 9 core  $\Lambda$ CDM parameters** are wired with UQFF integer-primitive expressions:

| $\Lambda$ CDM parameter | UQFF closure                 | Residual          |
|-------------------------|------------------------------|-------------------|
| H_0 (Planck)            | PAPER_1553                   | 0.015%            |
| H_0 (SH0ES)             | PAPER_1573                   | <b>EXACT</b>      |
| $\Omega_m$              | PAPER_1616                   | 0.143%            |
| $\Omega_\Lambda$        | PAPER_1617                   | 0.182%            |
| T_CMB                   | PAPER_1618                   | 0.064%            |
| Age (universe)          | PAPER_1619                   | 0.045%            |
| $\sigma_8$              | PAPER_1620                   | 0.253%            |
| n_s                     | (PAPER_1209GG_S650, earlier) | 0.072%            |
| z_recomb                | PAPER_1552                   | <b>EXACT 1090</b> |

Standard  $\Lambda$ CDM cosmology is now **essentially complete in UQFF** — every primary parameter measurable by Planck/WMAP/SDSS/DES collaborations has a UQFF integer-primitive form.

**NOT REPLACEMENT**

Empirical cosmology measures this constant directly. UQFF supplies a structural derivation from the integer primitives.

**Reference**

- Source: PAPER\_1209GG\_S644
- Related: PAPER\_1494-1615 (other session-2026-06-18 closures)
- Calculator dispatch: calculate\_paradox({"paradox": "omega\_lambda\_dark\_0\_685"})

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